

HGCAL Validation Results

Release Validation Campaigns

CMSSW_16_0_X

CMSSW_16_0_0_pre1_vs_CMSSW_15_1_0_pre6

CloseByParticleGun in CE-E Front section with 200 um

Reconstructed hits calibration;

ReconstructableEnergyOverCPenergy/h EoP CPene 300 calib fraction.png

Entries: New one is 44 but old one is 70

Graph: Old top point is much higher then new top point

ReconstructableEnergyOverCPenergy/h EoP CPene 100 calib fraction.png

Old values are zero

Std Dev: New value is also zero

ReconstructableEnergyOverCPenergy/h EoP CPene scint calib fraction.png

All valeus are zero

Simulated hits, digis, reconstructed hits validation

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Layer clusters

Efficiencies vs layer/globalEfficiencies.png

There is very little data and no overlap

Efficiencies vs layer/globalEfficiencies.png

Overlaps decrease as layer numbers increase.

Duplicates vs layer/globalEfficiencies.png

Empty

Duplicates vs layer/globalEfficiencies.png

Empty

FakeRate vs layer/globalEfficiencies.png

There is very little data and no overlap

FakeRate vs layer/globalEfficiencies.png

All value are zero

MergeRate vs layer/globalEfficiencies.png

Empty

MergeRate vs layer/globalEfficiencies.png

Empty

Tracksters

+

Tracksters vs SimTracksters from CPs Associatedby Hits

+

Tracksters vs SimTracksters from CPs Associatedby LayerClusters

+

Tracksters vs SimTracksters Associated by Hits

+

Tracksters vs SimTracksters Associated by LayerClusters

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